

**Topic: Section 1**

Place your test image detection results here.

Add a brief caption (2-3 sentences) describing how well you think the stop sign detection is working, noting any false positives or negatives, and what features in the image it might be picking up on or failing to pick up on.

The detector was able to detect stop signs in most of the test images. In the ones where the stop sign was smaller or cutt off, the detector would fail to detect or would send weird shaped boxed. There were some false positives in cluttered scenes so it has a hard time with those edges perhaps. It might be picking up on the color and hard edge detection, not too sure.

I predict: the first image will be a easy detection on the stop sign but I am not sure how the building roofs and cones might do. For the second image I think that there is gonna be a easy time not detecting since there isnt much that could come across as a stop sign.

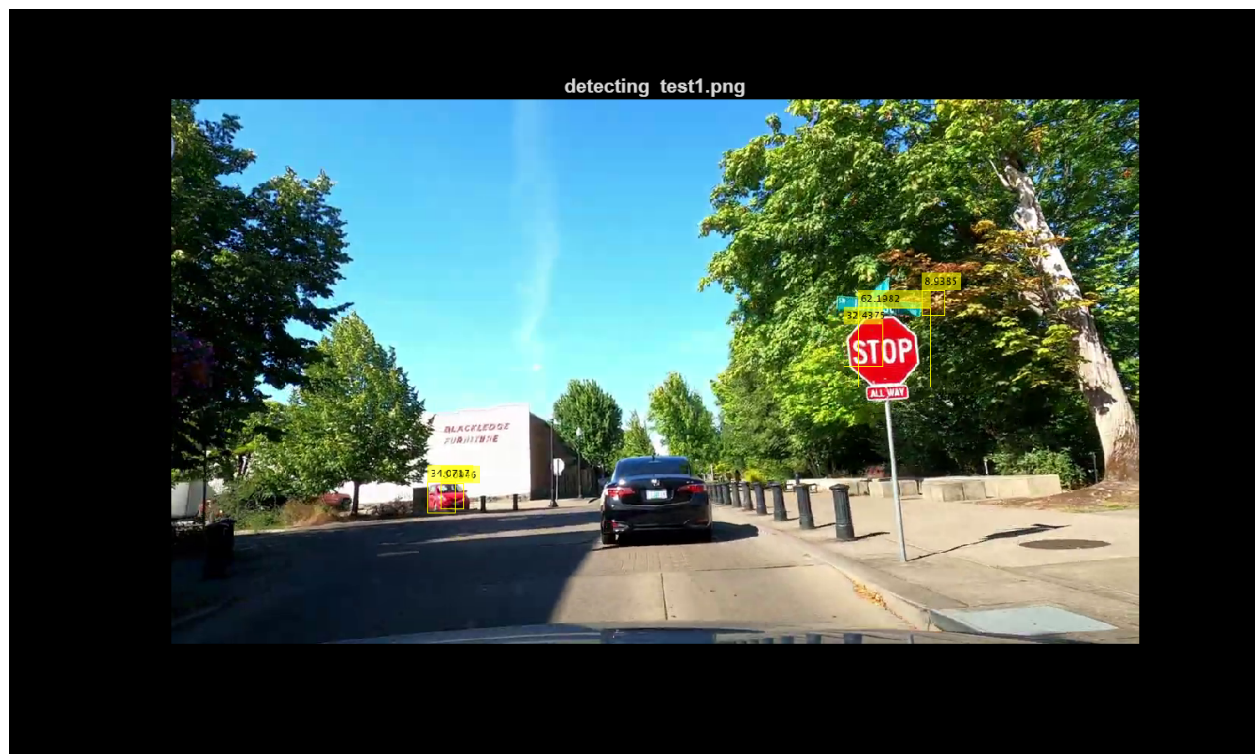


Figure 1: Vision Testing



Figure 2: Vision Testing

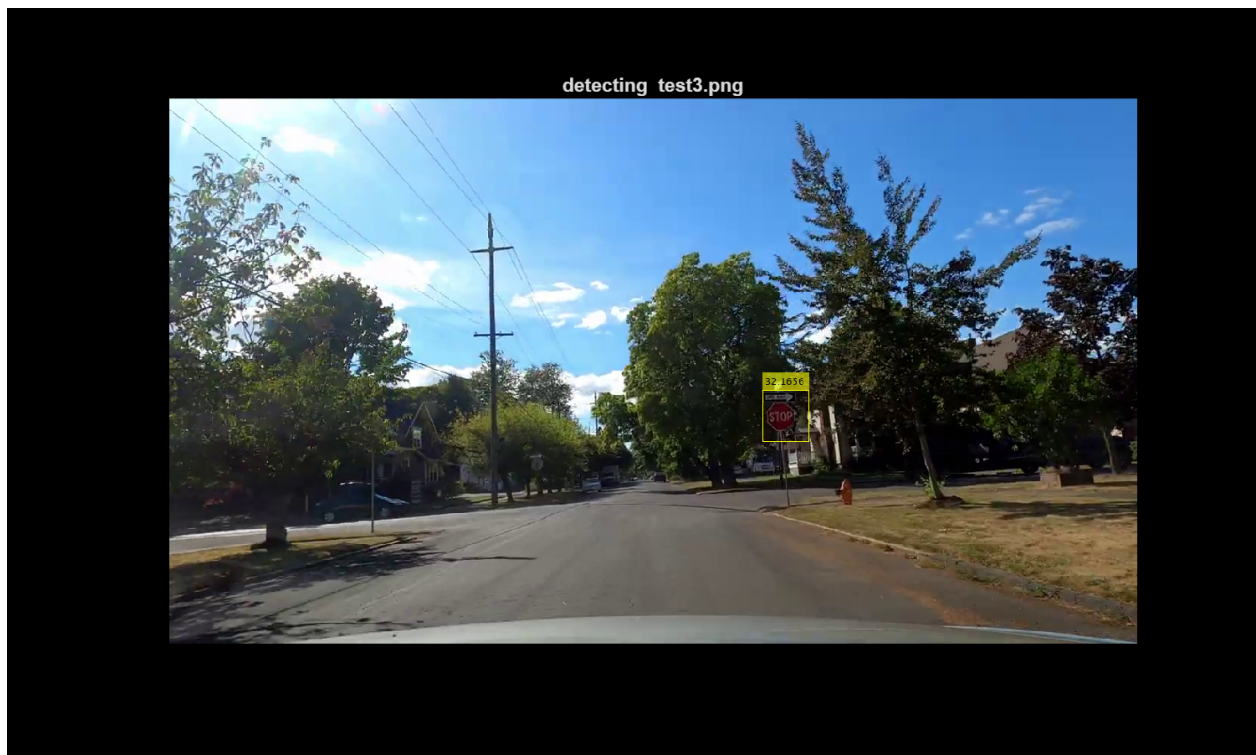


Figure 3: Vision Testing



Figure 4: Vision Testing

**Topic: Section 2**

Place your prediction image detection results here.

Add a brief caption (2-3 sentences) describing how well your predictions went and what discrepancies, if any, there were between your predictions and the actual results.

I predicted that figure 5 would be detected (kinda of correct) and that figure 6 would not be detected (not correct) just because of how obvious it seemed. After running it, I came to the conclusion that is might be assignnig a "weight" to how close hard boundaries are from one another to maybe account for the angles in an octagon sign. I didn't think it would be that sensitive to everything but it must be, the graininess must be pretty tough too on it also.



Figure 5: Prediction 1





Figure 6: Prediction 2

**Topic: Section 3**

Place your detection-tricking image detection results here.

Add a very brief caption (1-2 sentences) describing why you thought this image might trick the detector.

First, I wanted a grainy picture so I knew I would want one of my film folder. I also wanted something with more reds and sharp lines like the clustered regions would have. Thats it, nothing crazy.



Figure 7: Doggo face

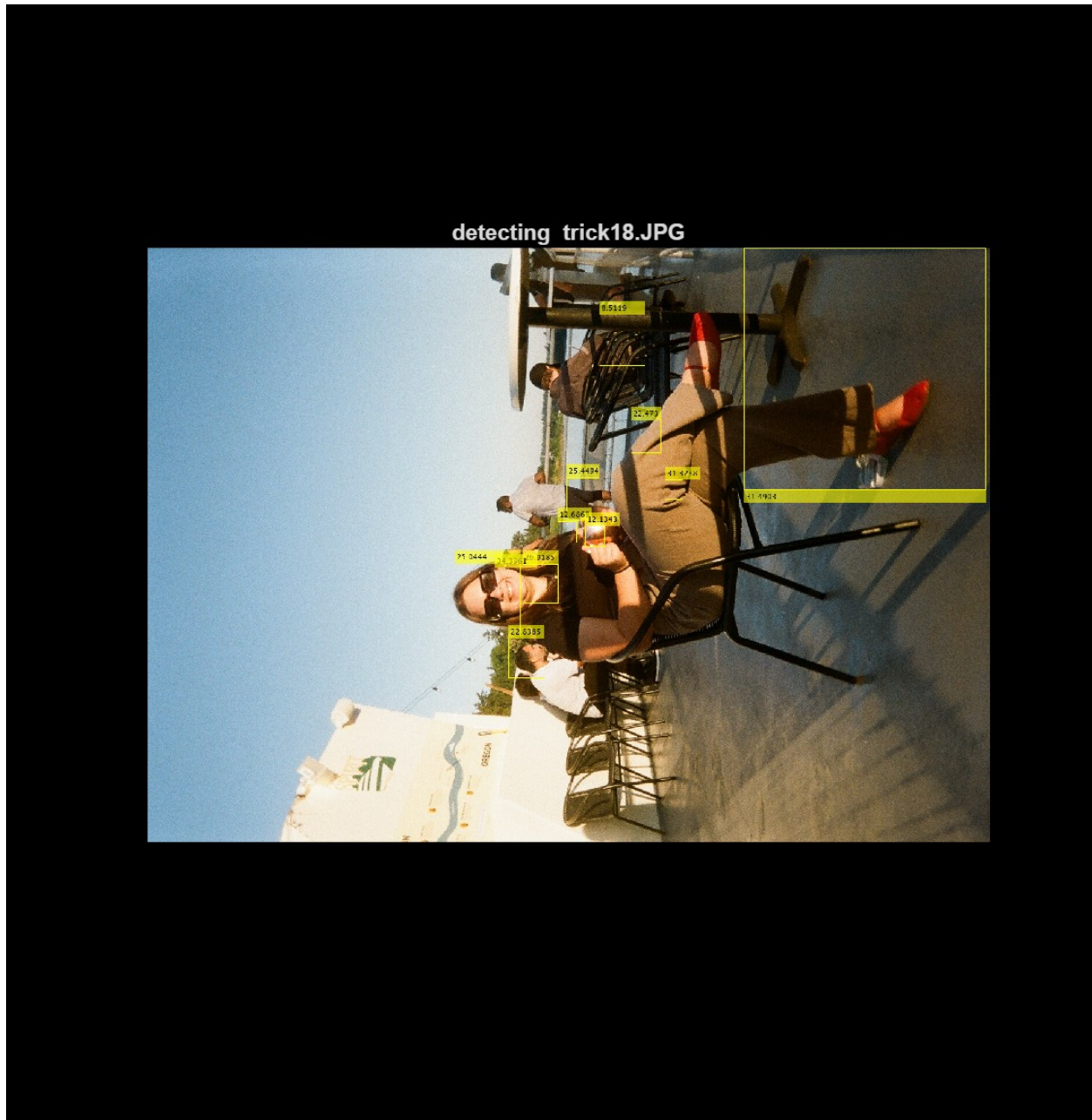


Figure 8: I felt pretty good about this one.

**Topic: Section 4**

Place your R-CNN image detection results here. Add a brief caption (2-3 sentences) comparing the effectiveness of the R-CNN compared to the ACF detector, as well as their computational efficiency (how long does the R-CNN take to train and run compared to the ACF detector?).

The rCNN seems to be more effective than the ACF, if you have the time for the compute. Training the rCNN took much much longer than the ACF but that is to be expected with the convolution and the loss calcs that happen. Like any  $\eta = \frac{\text{output}E}{\text{Inout}E}$ , it just matters if the ratio is enough for what you need.

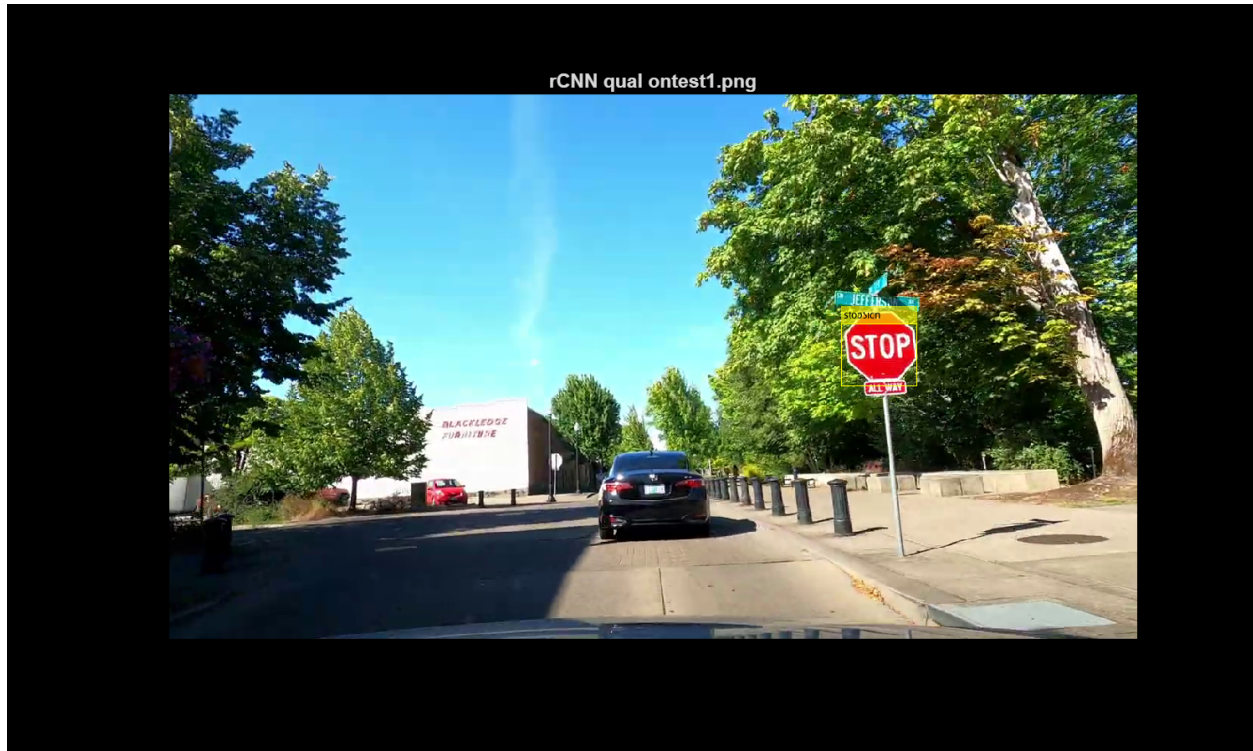


Figure 9: rCNN going through a qual on the test images.



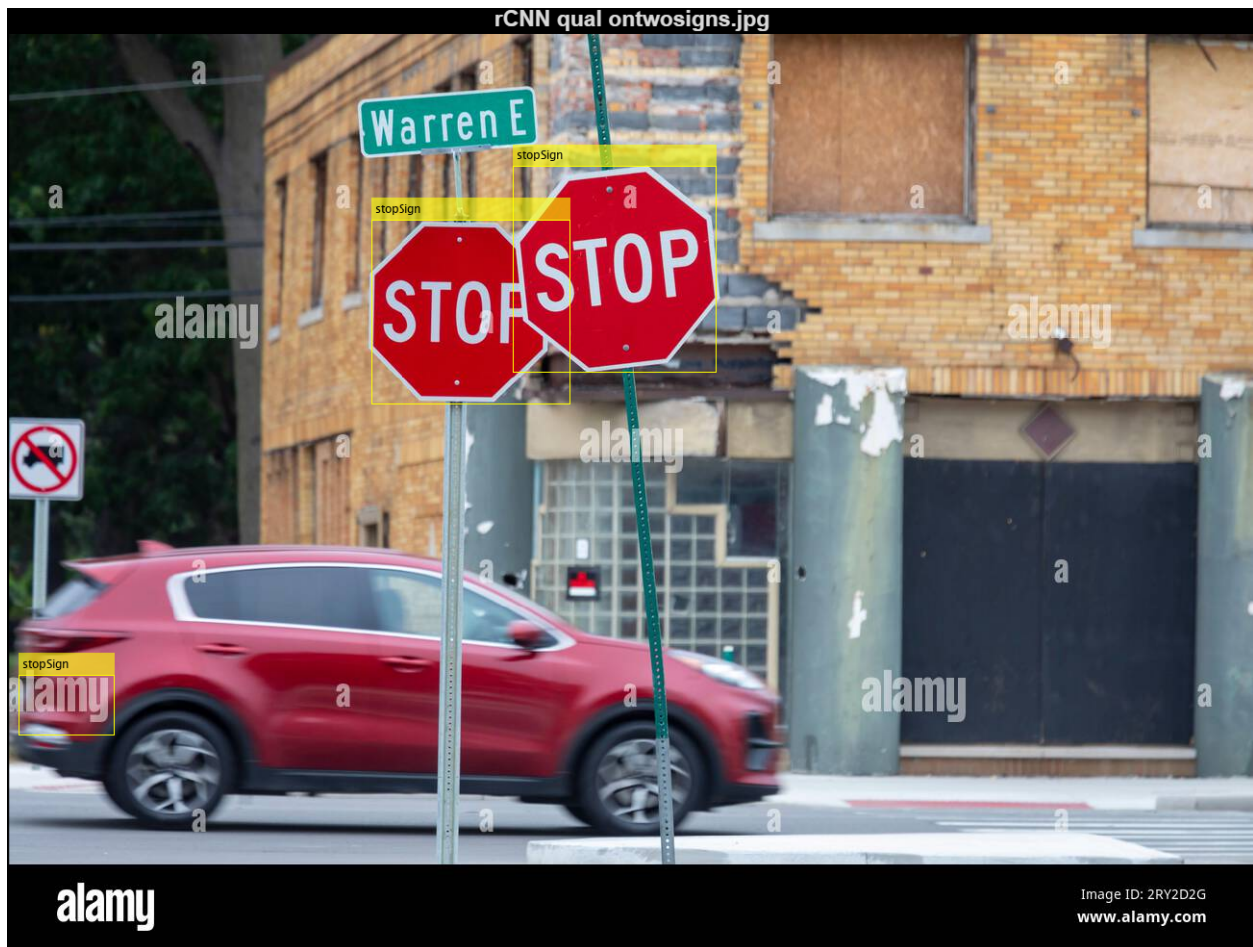


Figure 10: rCNN qual2

This one happend to flag the text on the bottom and I am wondering if that is due to it picking up on the white from the letters and the red from the car.

**Topic: Section 5**

Please list the following:

1. Any online sources you used and what they helped with. If you used any generative AI tools (e.g., ChatGPT, Gemini, Copilot, Claude, etc.), include a screenshot of your entire chatlog.
2. Who you worked with in the lab (excluding teachers) and what they helped with.

Online sources should be notated cleanly with the name and a hyperlink. “What they helped with” should focus on conciseness.

Examples: “Wikipedia helped me understand citations,” “John worked with me on Sign-off 2.” If the answer is “No online sources used.” or “No collaborators.”, please note this in the report for the corresponding item, so we know you considered both prompts.

<https://www.geeksforgeeks.org/computer-vision/how-does-r-cnn-work-for-object-detection/>

<https://www.mathworks.com/matlabcentral/answers/506371-trying-to-classify-images-with-a-cnn-but-getting-errors>

<https://www.mathworks.com/matlabcentral/answers/506412-error-with-classes-in-network-classification-layer>

<https://www.mathworks.com/help/deeplearning/ref/nnet.cnn.layergraph.replacelayer.html>

<https://youtu.be/zfiSAzpy9NM?si=q1H1rK0YmStaJfIO>

**Page 6:** Self-evaluation:

- (a) What did you expect to learn? I was hoping to learn what it meant to train and use a ML model. I had done something like that before so this was nice to see.
- (b) How would you rate your attendance and participation in class during this lab experience? Are any adjustments needed? It was good this last week. I got some larger assignments balanced out thankfully.
- (c) How well did you do at the lab assignment itself? You might include thinking about strengths and weaknesses of your performance in this response. Please also give yourself a grade (out of 4 possible points). I would give myself a 4 for this. I kept on top of everything including my attendance in lab.
- (d) What did you actually learn in the lab? (This might include a range of things, from subject matter and concrete skills to ways of knowing and working together with peers.) I learned that no one should ever set a training round for 10 epochs on their Microsoft surface. I also learned just how sensitive the more accessible vision algos are. rCNN is nice but it seems pretty heavy to use in a dynamic way. I don't really have experience with that but I would assume that there is a type of processing bottleneck if its something that is going to be used on a car or or where a robot needs to keep learning and updating. This was definitely my favorite lab of the term. I learned a lot about the convolution and Im pretty excited about it but like I said before, ill spare you the details.